



1. A shopkeeper sells half of the oranges he had plus 1 to the first customers. Then $\frac{1}{3}$ of the remaining plus 1 to his second customer and $\frac{1}{5}$ of the remaining plus 1 to his third customer. If he is left with 3 oranges at the end, then initially he had: **(Number System) NIMCET-2007**
(a)12 oranges (b)24 oranges
(c)36 oranges (d)none
2. A, B, D, E, F, G are consecutive integers between 1-10 not in same order satisfying the following conditions. D is 3 less than A, F is as much less than A as C is greater than D, B occupy central position. If $A = 6$, then the sum of E and G.**(Number System) NIMCET-2007**
(a)8 (b)12 (c)15 (d)none
3. A train starts from a station with certain passengers. At its first half, half of the passenger gets down. At the end, it is left with 360 passenger. How many passengers were there in the train initially: **(Number System) NIMCET-2007**
(a)3600 (b)4500 (c)3000 (d)none
4. The smallest number which when divided by 2, 3, 4, 5 leaves 1 as remainder in each case but, leaves no remainder when divided by 7: **(Number System) NIMCET-2007**
(a)231 (b)301 (c)331 (d)none
5. Three thieves stole in a bakery bread shop one by one. Each took with him half of the present bread plus half bread. If at last 3 breads are left. Then initially the bakery shop had breads: **(Number System) NIMCET-2007**
(a)15 (b)31 (c)24 (d)none
6. Which number should be subtracted from the highest four digit number to make it a perfect square: **(Number System) NIMCET-2007**
(a)197 (b)198 (c)159 (d)none
7. How many arrangements of four 0's, two 1's and two 2's are there in which the first 1 occurs before the first 2? **(Number System) NIMCET-2007**
(a)210 (b)420 (c)105 (d)None
8. If x and y are the two digits of the number 565xy such that this number is divisible by 80, then $x + y$ is equal to **(Number System) NIMCET-2008**
(a)2 (b)3 (c)8 (d)6
9. If both 7^2 and 3^3 are factors of the number $(a \cdot 11^3 \cdot 6^2 \cdot 3^{11})$, then what is the smallest possible value of a? **(Number System) NIMCET-2008**
(a)1323 (b)147 (c)21 (d)3087
10. From a height of 16 m, a ball fell down and each time it bounces half the distance back. What is the distance travelled? **(Number System) NIMCET-2008**
(a)45 m (b) ∞ (c)48m (d)24 m
11. There were a total of 10 bicycles and tricycles. If the total number of wheels was 24, how many tricycles were there? **(Number System) NIMCET-2008**
(a)4 (b)6 (c)8 (d)2
12. A certain street has 1000 buildings. A sign-maker is contracted to number the houses from 1 to 1000. How many zeroes will be needed? **(Number System) NIMCET-2008**
(a)128 (b)190 (c)181 (d)192
13. Of the three numbers, second is twice the first and is also thrice the third. If the average of three numbers is 44, the largest number is **(Number System-2008)**
(a)24 (b)36 (c)72 (d)108
14. An airline has a certain free luggage allowance and charges for excess luggage at a fixed rate per kg. Two passengers Raja and Rahim have 60 kg luggage between them, and are charged ₹ 1200 and ₹ 2400, respectively for excess luggage. Had the entire luggage belonged to one of them, the excess luggage charge would have been ₹ 5400. What is the weight of Rahim's luggage? **(Number System) NIMCET-2008**
(a)20 kg (b)25 kg
(c)30kg (d)35 kg
15. A group of 630 children is arranged in rows for a group photograph session. Each row contains three fewer children than the row in front of it. What numbers of rows is not possible? **(Number System) NIMCET-2008**
(a)3 (b)4 (c)5 (d)6
16. Sum of all three digit numbers (no digit being zero) having the property that all digits are perfect squares is **(Number System) NIMCET-2009**
(a)3108 (b)6216
(c)13986 (d)None of these
17. A teacher gave a student the task of adding 'N' natural numbers starting from 1. After a while, the student reported his result as 700. The teacher replied that his result was wrong. The student realized that he has added one number 24r twice by mistake. Find the sum of the digits of the number which the student had added twice. **(Number System) NIMCET-2009**
(a)5 (b)6 (c)7 (d)8
18. Divide Rs. 1074 (in whole Rs. having incremental amounts) into a number of bags so that I can ask for any amount between Rs. 1 and Rs. 1074 and you can give me the proper amount by selecting a certain number of these bags without opening them. What is the minimum number of bags you will require? **(Number System) NIMCET-2009**
(a)12 (b)10 (c)9 (d)11
19. The remainder when $x = 1! + 2! + 3! + 100!$ is divided by 240 is **(Number System) NIMCET-2009**
(a)153 (b)33 (c)73 (d)187
20. Using the digits 1, 5, 2, 8 all possible four digit numbers is between **(Number System) NIMCET-2009**
(a)10000 and 2000 (b)20000 and 50000
(c)50000 and 100000 (d)100000 and 150000
21. The sum of the numbers from 1 to 100, which are not divisible by 3 and 5, is **(Number System) NIMCET-2009**





- (a)2946 (b)2732 (c)2632 (d)2317
22. If all the 6's are replaced by 9's, then the algebraic sum of all the numbers from 1 to 100 (both inclusive), varies by **(Number System) NIMCET-2009**
(a)330 (b)333 (c)219 (d)279
23. When you reverse the digits of the number 13, the number increases by 18. How many other two-digit numbers increase by 18 when their digits are reversed? **(Number System) NIMCET-2009**
(a)5 (b)6 (c)7 (d)8
24. How many numbers from 1 to 100 are such each of which is divisible by 8 and whose at least one digit is 8 **(Divisibility) NIMCET-2010**
(a)Four (b)Zero
(c)Eight (d)Six
25. How many numbers between 1 to 1000 (both excluded) are both squares and cubes? **(Number System) NIMCET-2010**
(a)none (b)1 (c)2 (d)3
26. At a dance party a group of girls and boys exchange dances as follows : **(Number System) NIMCET-2011**
One boy dances with 5 girls, Second boy dances with 6 girls, and so on last boy dances with all girls. If b represents the number of boys and g represents the number of girls, then
(a) $b = g$ (b) $b = g / 5$
(c) $b = g - 4$ (d) $b = g - 5$
27. Suppose $X = 2^{100}$, $Y = 3^{100}$ and $Z = 4^{100}$. Exactly one of the following is true. Which is it? **(Number System) NIMCET-2011**
(a) $X + Y = Z$ (b) $X + Y < Z$
(c) $X + Y > Z$ (d) $XY = Z$
28. In a club, there are certain number of males and females. If 15 females are absent, then number of males will be half of females. If 45 males are absent, then female strength will be 5 times that of males. Number of males actually present is **(Number System) NIMCET-2012**
(a)45 (b)80
(c)105 (d)175
29. Let x, y and z be distinct integers, x and y are odd and positive and z is even and positive. Which one of the following statements cannot be true? **(Number System) NIMCET-2012**
(a) $(x-z)^2 y$ is even (b) $(x-z) y^2$ is odd
(c) $(x-z) y$ is odd (d) $(x-y)^2 z$ is even
30. Three men A, B and C plays cards. If one loses the game he has to give ₹ 3. If he wins the game he will gain ₹ 3 each from the other two losers. If A has won 3 games, B loses ₹ 3, C wins ₹ 12, then the total number of games played is **(Number System) NIMCET-2012**
(a)12 (b)21
(c)20 (d)6
31. In a certain school, the number of student in each section was 24. After admitting some students, three new sections

- have been started and now there are 16 sections with 21 students in each. What is the number of newly admitted students? **(Number System) NIMCET-2014**
(a)14 (b)24 (c)16 (d)26
32. The nine alphabets L, M, N, O, P, Q, R, S and T are assigned to nine integers 1 to 9 not necessarily in the same order 4 is assigned to P. The difference between P and T is 5. The difference between N and T is 3. What is the integer assigned to N? **(Number System) NIMCET-2014**
(a)7 (b)6 (c)5 (d)4
33. Ajith is three times older than Babita. Chetu is half the age of Das. Babita is older than Chetu. Which of the following additional information is needed to estimate the age of Ajith? **(Number System) NIMCET-2014**
I. Chetu is 10 yr old.
II. Both Babita and Das are older than Chetu by the same number of years.
(a) Only I (b) Only II
(c) I and II (d) None of these
34. Which letter is the word CYBERNETICS occupies the same position as it does in the English alphabet? **(Number Series) NIMCET-2015**
(a)C (b)E
(c)I (d)T
35. The remainder when 2^{31} is divided by 5 is **(Number System) NIMCET-2015**
(a)1 (b)2
(c)3 (d)4
36. How many 3- digit numbers divisible by 5, can be formed using the digits 2, 3, 5, 6, 7 and 9, without repetition of digits? **(Number System) NIMCET-2015**
(a)216 (b)20
(c)17 (d)18
37. Using only 2, 5, 10, 25 and 50 paise coins, what is the smallest number of coins required to pay exactly 78 paise, 69 paise and Rs.1.01 to three different persons? **(Number System) NIMCET-2015**
(a)19 (b)20
(c)17 (d)18
38. A drawer contains 10 black and 10 brown socks which are all mixed up. What is the smallest number of socks to be taken from the drawer to decided without seeing them, to be sure that there is atleast one pair of socks of the same colour? **(Number System) NIMCET-2015**
(a)11 (b)10
(c)3 (d) Cannot be determined
39. If all the 6's are replaced by 9's then the algebraic sum of all the numbers from 1 to 100 (both inclusive) varies by **(Number System) NIMCET-2016**
(a) 333 (b) 300 (c) 279 (d) 330
40. 405 sweets were distributed equally among a group of children such that the number of sweet received by each





- child is one fifth of the number of children. The number of children in the group is **(Number System) NIMCET-2016**
(a) 45 (b) 9 (c) 21 (d) 15
41. A bus starts from its depot filled to seating capacity. It stops at a point A where $\frac{1}{6}$ th of the passengers alight and 10 board the bus. At point B, $\frac{1}{5}$ th of the passengers alight and boards the bus. At point C which is the last stop, all the 55 passengers alight. The capacity of the bus is **(Number System) NIMCET-2016**
(a) 55 (b) 60 (c) 66 (d) none
42. A Group of 630 children are seated in n rows for a group photo session. Each row contains three less children than the row in front of it. Which one of the following number of rows is not possible? **(Number System) NIMCET-2016**
(a) 3 (b) 4 (c) 5 (d) 6
43. What is the largest number of positive integers to be picked up randomly so that the sum of difference of any two of the chosen numbers is divisible by 10? **(Number System) NIMCET-2017**
(a) 2 (b) 5 (c) 7 (d) 10
44. A caterpillar crawls up a pole of 75 inches high standing from the ground. Each day it crawls up 5 inches and each night it slides down 4 inches. When will it reach the top of the pole? **(Number System) NIMCET-2017**
(a) At the end of 70 days (b) At the end of 71 days
(c) At the end of 72 days (d) At the end of 73 days
45. How many numbers between 1 and 1000 are divisible by all numbers 2, 3, 4, 5 and 6? **(Number system) NIMCET-2018**
(a) 16 (b) 32 (c) 17 (d) 33
46. There are 8436 steel balls, each with a radius of 1cm, stacked in a pile, with 1 ball on top, 3 balls in second layer, 6 in the third layer, 10 in the fourth and so on. the number of horizontal layer, in the pile is **(Number system) NIMCET-2018**
(a) 34 (b) 38 (c) 36 (d) 32
47. The last digit of the number obtained by multiplying the numbers **(number System) NIMCET-2018**
 $81 \times 82 \times 83 \times 84 \times 85 \times 86 \times 87 \times 88 \times 89$ will be
(a) 0 (b) 9 (c) 7 (d) 2
48. Let n be the number of different 5 digit numbers, divisible by 4 that can be formed with the digits 1, 2, 3, 4, 5 and 6, with no digit being repeated. What is the value of n ? **(Number System) NIMCET-2018**
(a) 144 (b) 168 (c) 192 (d) 222
49. Let S be the set of integers X such that
i. $100 \leq x \leq 200$
ii. x is odd and
iii. x is divisible by 3 but not by 7
How many elements does S contain? **(Number System) NIMCET-2018**
(a) 16 (b) 12 (c) 11 (d) 13
50. Two bus tickets from city A to B and three tickets from city A to C cost Rs.77, but three tickets from city A to B and two tickets from city A to C cost Rs.73. What are the fares for cities B and C from A? **(Number System) NIMCET-2018**
(a) Rs. 4, Rs. 23 (b) Rs. 13, Rs. 17
(c) Rs. 15, Rs. 14 (d) Rs. 17, Rs. 13
51. A caterpillar crawls up a pole of 75 inches high, starting from the ground. Each day, it crawls up 5 inches and each night it slides down 4 inches. When will it reach the top of the pole? **(Number System) NIMCET-2018**
(a) End of 70 days (b) End of 71 days
(c) End of 72 days (d) End of 73 days
52. Some friends planned to contribute equally to jointly buy a CD player. However, two of them decided to withdraw at the last minute. As a result, each of the others had to shell out one rupee more than what they had planned for. If the price (in ₹) of the CD player is an integer between 1000 and 1100, find the number of friends who actually contributed? **(Number System) NIMCET-2019**
(a) 44 (b) 23 (c) 21 (d) 46
53. How many positive numbers less than 10000 are such that the product of their digits is 210? **(Number System) NIMCET-2019**
(a) 36 (b) 42 (c) 48 (d) 54
54. Each of A, B and C is a different digit among 1 to 9. How many different values of the sum of A, B and C are possible, if $ABA \times AA = ACCA$? **(Number System) NIMCET-2019**
(a) 1 (b) 3 (c) 7 (d) 8
55. How many 4-digit numbers that can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated? **(Number System) NIMCET-2019**
(a) 216 (b) 60 (c) 24 (d) 25
56. Using only 2, 5, 10, 25 and 50 paise coins, the smallest number of coins required to pay exactly 79 paise, 66 and ₹1.01 to three different persons is **(Number System) NIMCET-2019**
(a) 17 (b) 20 (c) 19 (d) 18
57. A total of 324 notes comprising of ₹20 and ₹ 50 denomination make a sum of ₹ 12450. The number of ₹ 20 notes is **(Number System) NIMCET-2020**
(a) 200 (b) 144 (c) 125 (d) 110
58. Choose the correct option for the remainder when $X = 1! + 2! + 3! + \dots + 100!$ is divided by 24. **(Number System) NIMCET-2021**
(a) 9 (b) 11 (c) 152 (d) 13
59. Two friends A and B were standing at the diagonally opposite corners of a rectangular plot whose perimeter is 100m. A first walked x m along the length of the plot towards East and then y m towards the south. B walked x m along breadth towards North and then y m towards West. At the end their walks, A and B were standing at the diagonally opposite corners of a smaller rectangular plot



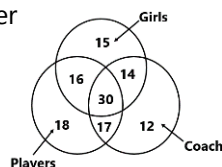
whose perimeter is 40m. How much distance did A walk?

(Number System) NIMCET-2023

- (a)15 (b)25 (c)40 (d)50

60. Study the diagram and answer

The question that follows:



How many players are there who are girls but not a coach?

(Number System) NIMCET-2023

- (a)16 (b)31 (c)15 (d)61

61. 9. A cat climbs a 21- meter pole. In the first minute it climbs 3 meter and in the second minute it descends one meter. In how minutes the cat would reach the top of the pole?

(Number System) NIMCET-2024

- (a) 18 min (b) 21 min
(c) 19 min (d) 20 min

ANSWER

1	d	2	a	3	d	4	b	5	b
6	*	7	a	8	d	9	b	10	c
11	a	12	d	13	c	14	d	15	d
16	a	17	c	18	d	19	a	20	d
21	c	22	a	23	b	24	a	25	c
26	c	27	b	28	b	29	a	30	a
31	b	32	b	33	c	34	c	35	c
36	b	37	a	38	c	39	d	40	a
41	c	42	d	43	b	44	b	45	a
46	c	47	a	48	c	49	d	50	b
51	b	52	a	53	d	54	d	55	b
56	d	57	c	58	a	59	a	60	a
61.	c								

SOLUTION

1. (d) Let shopkeeper has total = x orange

$\left(\frac{x}{2} + 1\right)$ = first customer

second customer = $\frac{1}{3}\{x - (\frac{x}{2} + 1)\} + 1 = \frac{x+4}{6}$

third customer = $\frac{1}{5}\{x - [(\frac{x}{2} + 1) + \frac{x+4}{6}]\} + 1 = \frac{x+10}{15}$

Finally he have 3 left

finally he have $3 - \left[\left(\frac{x}{2} + 1\right) + \left(\frac{x+4}{6}\right) + \left(\frac{x+10}{15}\right)\right] = 3$

$$x - \frac{15x+30+5x+20+2x+20}{30} = 3$$

$$\frac{30x-22x-70}{30}$$

$$8x = 90 + 70$$

$$8x = 160$$

$$X = 20$$

2. (a)

D is 3 less than A, and A = 6.

$$D = A - 3$$

$$D = 6 - 3$$

$$D = 3$$

A, B, D, E, F, G are consecutive integers.

B is the middle term, so the sequence is of the for D, F, E, B, C, A, G or G, A, C, B, E, F, D

Since A = 6 and D = 3, the consecutive intetgers are 3,4,5,6,7,8,9 or 9,8,7,6,5,4,3

B is the middle term, so B = 6.

F is as much less than A as C is greater than D.

$$A - F = C - D$$

Since A, B, D, E, F, G are consecutive and B is the middle term, the sequence 3,4,5,6,7,8,9

$$F = 4$$

$$6 - 4 = C - 3$$

$$C = 5$$

So

$$D = 3, F = 4, C = 5, A = 6, B = 6$$

$$E = 7, G = 9$$

$$\text{The sum of E and G} = 7 + 9 = 16$$

3. (d)

Let x be the initial number of passengers.

After half get off, $\frac{x}{2}$ passengers remain.

This remaining number is equal to 360.

$$\frac{x}{2} = 360$$

The initial number of passengers in the train was 720.

4. (b)

(2, 3, 4, 5)

$$\text{LCM} \Rightarrow 60 + 1 = 61$$

But it is totally divided by 7

$$\frac{60x+1}{7} \quad x = 5.$$

then put the value of x = 60×5 + 1 = 301

5. (b) let total breads = x

first the stole = $\frac{x}{2} + \frac{1}{2}$ breads

$$\text{second thief } \frac{x - (\frac{x}{2} + \frac{1}{2})}{2} + \frac{1}{2} = \frac{x/2+1/2+1}{2} = \frac{x+1}{4}$$

$$\text{third thief } = \frac{(\frac{x}{2} + \frac{1}{2}) - (\frac{x+1}{4})}{2} + \frac{1}{2} = \frac{2x-2-x-1}{8} + \frac{1}{2} = \frac{x-3+4}{8} = \frac{x+1}{8}$$

$$d \left(t = x - \left(\frac{x}{2} + \frac{1}{2} + \frac{x+1}{4} + \frac{x+1}{8} \right) \right) = 3$$

$$\frac{8x - (4x+4+2x+2+x+1)}{8} = 3$$

$$\frac{8x-7x-7}{8} = 3$$

$$x = 24 + 7 = 31 \text{ (option-b)}$$

6. Finding square root of 9999 by long

99	
9	9999
+9	81
189	1899
	1701
	198





So 198 should be subtracted from the highest 4 digit number, to make it perfect square.

7. (a) Number of zeros = 4

Number of ones = 2

Number of twos = 2

Total no. = 8

Remaining two are fixed in order total arrangements =

$$\frac{{}^8P_6}{2! 2! 4!} = 210$$

8. (d)

565xy will be divisible by 80, if $x = 6$ and $y = 0$

$$\Rightarrow x + y = 6$$

9. (b)

7^2 and 3^3 will be factor of

(a. $11^3 \cdot 6^3 \cdot 13^{11}$) If $a = 7^2 \cdot 3 = 147$.

10. (c)

Required distance

$$= 16 + 2 \left[16 \times \frac{1}{2} + 16 \times \frac{1}{2^2} + 16 \times \frac{1}{2^3} + 16 \times \frac{1}{2^4} + \dots \right]$$

$$= 16 + 2 \frac{16 \times \frac{1}{2}}{1 - \frac{1}{2}} = 48 \text{ m}$$

11. (a)

If x tricycles are there, then there will be $(10 - x)$ bicycles.

$$3x + 2(10 - x) = 24$$

$$x + 20 = 24$$

$$\Rightarrow x = 4$$

12. (d)

0 9 Case for two digit numbers.

↑

0

↑ - ↑ $9 \times 10 = 0$ cases

9 For three digit numbers.

0

↑↑ — $9 \times 10 = 0$ cases

In 1000 there is 3 O's

$$\text{Total O's} = 9 + 90 + 90 + 3 = 192$$

13. (c)

Second number is the largest. Let it be x , then first number is $x/2$ and third number is $x/3$.

So, their average is

$$\frac{x + \frac{x}{2} + \frac{x}{3}}{2} = 44$$

$$\Rightarrow \frac{11x}{18} = 44$$

$$\Rightarrow x = \frac{44 \times 18}{11} = 72$$

14. (d)

If free luggage is y kg, then Raja has x kg, and Rahim has $2x$ kg extra luggage.

Then, $2y + 3x = 60$

If $3x$ kg extra luggage cost 3600, then 5400 will be cost

$$\begin{aligned} \text{of} \\ \frac{5400}{3600} \times 3x &= \frac{9}{2}x \\ \Rightarrow y + 3x &= \frac{9}{2}x \end{aligned}$$

$$\Rightarrow y = \frac{3}{2}x$$

$$\Rightarrow x = 10 \text{ kg}$$

and $y = 15 \text{ kg}$

So, Rahim's luggage is $y + 2x = 35 \text{ kg}$

15. (d)

If x is the number of rows, then if number of children in last row is y , then

$$y + (y + 3) + (y + 6) + \dots + \{y + 3(x - 1)\} = 630$$

$$\Rightarrow xy + \frac{3(x-1)}{2} 630$$

$$x = 3 \Rightarrow 3y + 9 = 630 \text{ (possible, } y = 207)$$

$$x = 4 \Rightarrow 4y + 18 = 630 \text{ (possible, } y = 153)$$

$$x = 5 \Rightarrow 5y + 30 = 630 \text{ (possible, } y = 120)$$

$$x = 6 \Rightarrow 6y + 45 = 630 \text{ (impossible)}$$

16. (a) Digits being perfect square and non-zero include 1, 4 and 9.

At each place digit appear $2! - 2$ times, so sum of all the three digit numbers

$$= (1 + 4 + 9) \times 2!(111)$$

$$= 28(111) = 3108$$

17. (c) As $1 + 2 + \dots + N = \frac{N(N+1)}{2}$

$\frac{N(N+1)}{2}$ just less than 700 is obtained for $N=36$ and $N(N+1)/2$ equals 666.

Thus, number $700 - 666 = 34$ was added twice. Hence, sum of digits = $3 + 4 = 7$

18. (d)

Amounts required in the bag will be powers of 2 i.e., 1, 2, 4, 8, 16, 32, 64, 128, 256, and 512.

Thus, minimum number of bags required = 11

19. (a)

As $6! = 720$, so numbers from 61 onwards is divisible by 240.

Hence, remainder is

$$1! + 2! + 3! + 4! + 5!$$

$$= 1 + 2 + 6 + 24 + 120 = 153$$

20. (d)

Sum of the numbers will be

$$(1 + 5 + 2 + 8) 3! (1111) = 96(1111) = 106656$$

which lies between 100000 and 150000

21. (c)

Required sum

$$= (1 + 2 + \dots + 100) - (3 + 6 + \dots + 99) - (5 + 10 + \dots + 100) + (15 + 30 + \dots + 90)$$

$$= 5050 - \frac{3 \times 33 \times 34}{2} - \frac{5 \times 20 \times 21}{2} + 15 \times 21$$

$$= 5050 - 1683 - 105 + 315 = 2632$$

22. (a)

In numbers from 1 to 100, 6 appears 10 times at unit place and also 10 times at ten's place.

Thus, by replacing 6 by 9, variation in the sum is

$$3 \times 10 + 3 \times 10 \times 10 = 330$$

23. (b)

Let the required numbers be xy .

$$\text{Then, } (yx) - (xy) = 18$$

$$10y + x - 10x - y = 18$$

$$\Rightarrow 9(y - x) = 18$$

$$\Rightarrow y - x = 2$$

Hence, possible numbers are 24, 35, 46, 57, 68, 79 besides 13.





24. (a) Number from 1 to 8 which is divisible by 8 and whose at least one digit is 8 are (8, 48, 80, 88)
total number of digit = 4

25. (c)
Perfect squares and cubes are = 2

26. (c) Suppose number of boys = b
1st boy dances with 5 girls ($1 + 4 = 5$)
2nd boy dances with 6 girls ($2 + 4 = 6$)
3rd boy dances with 7 girls ($3 + 4 = 7$)
 $\therefore$ bth boy dances with $(b + 4)$ girls
Hence, $g = b + 4 \Rightarrow b = g - 4$

27. (b) Suppose, $2^2 + 3^2 < 4^2$ ($\because 4 + 9 < 16$)
 $\Rightarrow 2^3 + 3^3 < 4^3$ ($\because 8 + 27 < 64$)
 $\Rightarrow 2^4 + 3^4 < 4^4$ ($\because 16 + 81 < 256$)

So, solving the pattern we conclude that for higher value

$$n > 1$$

$$2^n + 3^n < 4^n$$

$$\text{Then, } X + Y < Z$$

28. (b) Let x and y be the certain number of males and females.

Then, by condition,

$$x = \frac{1}{2}(y - 15)$$

$$\Rightarrow 2x = y - 15$$

$$\Rightarrow 2x - y = -15$$

$$5(x - 45) = y$$

$$5x - y = 225$$

And On subtracting Eq. (i) from Eq. (ii), we get

$$3x = 240$$

$$\Rightarrow x = 80$$

$$\therefore \text{Number of males} = 80$$

29. (a) x, y, z are distinct integers.
and x and y are odd positive integers and z is even positive integers.

Then, $(x - z) = \text{Odd number}$

$$(x - z)^2 = \text{Odd positive number}$$

$$\text{and } (x - z)^2 \cdot y = \text{Odd} \times \text{Even} = \text{even}$$

30. (a)

Total money lost by B: ₹3.

Total money won by C: ₹12.

Total money won by A: let's calculate.

From A's 3 wins: $3 \times 6 = ₹18$.

But A might have lost money in other games.

If total games are G, A has lost $(G - 3)$ games, losing ₹3 each,

so total loss: $3 \times (G - 3)$.

Therefore, A's net gain is

$$18 - 3 \times (G - 3) = 27 - 3G.$$

For the total money to be conserved, total net gains should be zero:

$$\text{A's net gain} + \text{B's net gain} + \text{C's net gain} = 0.$$

Which is:

$$(27 - 3G) + (-3) + 12 = 0$$

$$27 - 3G + 9 = 0$$

$$36 - 3G = 0$$

$$G = 12.$$

31. (b) Total number of students when there were 24 students in each of

$$13 \text{ sections} = 13 \times 24 = 312$$

Total number of students when three new section were added with 21 students in each of 16 sections

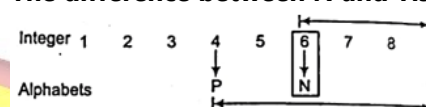
$$= 21 \times 16 = 336$$

$$\text{Number of newly admitted students} = 336 - 312 = 24$$

32. (b) On the basis of the given information, the integer assigned to N is 6.

This is explained below

The difference between N and T is 3



The difference between P and T is 5

33. (c) Let the ages of Ajith, Babita, Chetu, Das are A, B, C, D yr. Age of Ajith is three times that of Babita i.e. $A = 3B$

Chetu is half the age of Das i.e. $C = \frac{D}{2}$

Babita is older than Chetu i.e. $B > C$.

Statement I Chetu is 10 yr old. This statement is necessary to find the age of Ajith.

Statement II This statement is again necessary to find the age of Ajith.

Hence, both statements I and II are necessary to estimate the age of Ajith.

34. (c) In the word CYBERNETICS, I occupies the same position as it does in the English alphabet. The place value of 'I' is 9.

35. (c) Here, 2^{31} is divided by 5

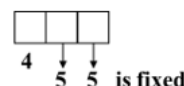
$$2^{31} = 2^{30} \times 2^1$$

$$= (2^2)^{15} \times 2$$

$$= 4^{15} \times 2$$

$$= (-1 + 5) \times 2 = 4 \times 2 = 8 \div 5 = 3$$

36. (b) Here



So, total 3 digits number divisible by 5

$$= 5 \times 4 \times 1 = 20$$

37. (a) Here,

We have, to find LCM of 78 paise, 69 paise and 101 paise.

And,

$$78 = 1 \times 50 + 2 \times 10 + 2 \times 4 = 5 \text{ coins}$$

$$69 = 1 \times 50 + 1 \times 10 + 1 \times 5 + 2 \times 2 = 5$$

$$101 = 1 \times 101 = 1 \times 50 + 4 \times 10 + 1 \times 5 + 3 \times 2 = 9$$

$$\text{So, LCM} = 2 \times 3 \times 13 \times 23 \times 101 = 181194$$

38. (c) Even if first 2 are different colours, 3rd one has to be black or brown. So, the smallest number of socks will be 3.



- 39. (d)** There are 19 numbers which has 6 as a digit when unit digit 6 is replaced by 9, then sum of numbers will be increase by $= 3 \times 10 = 30$
From 60 to 69 there are 10 numbers which have ten's digit 6.

Here, the increase will be $30 \times 10 = 300$

$$\therefore \text{Total increase} = 30 + 300 = 330$$

- 40. (a)** Let the number of children be x .

$$\therefore x \times \frac{x}{5} = 405$$

$$\Rightarrow x^2 = 5 \times 405 = 2025$$

$$\Rightarrow x = \sqrt{2025} = 45$$

- 41. (c)** At C, number of passenger = 55

$$\therefore \text{Total number of passenger at B} = \frac{5}{4} \times (55 - 3) \\ = \frac{5}{4} \times 20 = 65$$

$\therefore$ Total number of passengers at

$$A = \frac{6}{5} \times (65 - 10)$$

$$= \frac{6}{5} \times 55 = 66$$

- 42. (d)** We can check it through option.

- 43. (b)** Let largest number is P

and any two choosen numbers x and y .

According to the question $P + x - y$ is divisible by 10.

So,

P is largest then

$$P + x - y \geq 10$$

$$x - y \leq P$$

These inequalities can be written in equation form

and

$$P + x - y = 10$$

$$x - y = P$$

On subtracting eq. (iv) by (iii), we get

$$P + x - y - x + y = 10 - P$$

$$\Rightarrow P = 10 - P$$

$$\Rightarrow 2P = 10$$

$$P = \frac{10}{2} = 5$$

- 44. (b)** Caterpillar crawls upward $(5 - 4)$, 1 inch in a day.

Upto 70th day, caterpillar crawls upward 70 inches.

On 71st day, caterpillar crawls upward 75 cm at the end of the day.

- 45. (a)** LCM of 2,3,4,5 and 6 = 60

Hence, Number between 1 and 1000 which are divisible by 60 = 16

- 46. (c)** Number of balls in 1st layer = 1

Number of balls in 2nd layer = $3 = 1 + 2$

Number of balls in 3rd layer = $6 = 1 + 2 + 3$

Number of balls in 4th layer = $10 = 1 + 2 + 3 + 4$

$$\text{Now, the } n\text{th term} = \frac{n(n+1)}{2} = \frac{n^2}{2} + \frac{n}{2}$$

$$\text{Sum of } n \text{ terms of series} = \frac{1}{2} [\Sigma n^2 + \Sigma n]$$

$$= \frac{1}{2} \left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right] = \frac{n(n+1)(n+2)}{6}$$

$$\text{Now, } \frac{n(n+1)(n+2)}{6} = 8436$$

$$\text{or } n(n+1)(n+2) = 50616$$

The above condition is satisfied only for $n = 36$

- 47. (a)** The last digit can be obtained by multiplying the last digits of the given numbers.

Here, 82×85 , i.e. $2 \times 5 = 10$

Hence, the last digit would be 0.

- 48. (c)** For a number to be divisible by 4 the last two digits must be divisible by 4.

$\therefore$ The possible cases are

— — — 12		— — — 16
— — — 24		— — — 32
— — — 36		— — — 52
— — — 56		— — — 64

$$\therefore \text{Value of } n = 4 \times 3 \times 2 \times 1 \times 8 = 192$$

- 49. (d)** First number divisible by 3 between 100 and 200 = 102

Last number divisible by 3 between 100 and 200 = 198

Total numbers divisible by 3 between 100 and 200 = 33

Out of which 105, 126, 147, 168 and 189 are divisible by 7 and from remaining numbers

102, 108, 114, 120, 132, 138

144, 150, 156, 162, 174, 180, 186, 192 and 198 are even numbers.

$$\therefore x = 33 - 20 = 13$$

- 50. (b)** Let the fare of 1 ticket from city A to B be ₹ x and the fare of 1 ticket from city A to C be ₹ y

According to the question,

$$2x + 3y = 77$$

$$3x + 2y = 73$$

On solving Eqs. (i) and (ii), we get

$$x = 13, y = 17$$

$\therefore$ The fares for cities B and C from A are ₹ 13 and ₹ 17, respectively.

- 51. (b)** A caterpillar crawls up 5 inches each day and slides down 4 inches each night.

$\therefore$ In 1 day it crawls up 1 inch.

Hence, in 70 days it crawls up 70 inches.

At the end of 71 days it will crawl up 5 inches again.

Hence, at the end of 71 days it will reach the top of the pole.

- 52. (a)** Let the cost of CD be Rs. y

and total number of friends be x .

Now, each of them has to pay Rs. $\frac{y}{x}$.

Now, after withdrawing 2 friends the remaining has to pay $\frac{y}{x-2}$

Now, according to given condition,

$$\frac{y}{x-2} - \frac{y}{x} = 1$$

$$\Rightarrow y = \frac{x(x-2)}{2}$$

Now, given $1000 < y < 1100$



$$x = 46$$

So, number of friends who actually contributed is

$$\Rightarrow 46 - 2 = 44$$

53. (d) $210 = 1 \times 2 \times 3 \times 5 \times 7$

$= 1 \times 6 \times 5 \times 7$ (only 2×3 makes the single digit 6) so, four digit numbers with combinations of the digits {1, 6, 5, 7} and {2, 3, 5, 7} and three digit numbers with combinations of digits {6, 5, 7} will have the product of their digits equal to 210. Number using 1, 6, 5, 74! = 24

Number using 2, 3, 5, 74! = 24

Number using 6, 5, 7 = 3! = 6

Total numbers $24 + 24 + 6 = 54$

54. (d) $ABA \times AA = ACCA$

different value for A, B and C

A	B	C	sum
1	1	2	4
1	2	3	6
1	3	4	8
1	4	5	10
1	5	6	12
1	6	7	14
1	7	8	16
1	8	9	18

There are 8 values that satisfies the Eq. (i). Hence, 8 different values of sum A, B and C are possible.

55. (b) Number of required numbers $= 5 \times 4 \times 3 \times 1 = 60$

56. (d) For 1.01 paise,

$$1 \times 50 + 1 \times 25 + 2 \times 10 + 3 \times 2 = 7 \text{ coins}$$

For 66 paise,

$$1 \times 50 + 1 \times 10 + 3 \times 2 = 5 \text{ coins}$$

For 79 paise,

$$1 \times 50 + 2 \times 10 + 1 \times 5 + 2 \times 2 = 6 \text{ coins}$$

$\therefore$ Total smallest number of coins = $7 + 5 + 6 = 18$ coins

57. (c) Given, 324 notes comprised of Rs. 20 and Rs. 50 total sum is Rs. 12450.

Let the number of Rs. 20 notes = x

The number of Rs. 50 notes = y

According to the question,

$$x + y = 324 \quad \dots\dots\dots (i)$$

$$20x + 50y = 12450 \quad \dots\dots\dots (ii)$$

Solving Eqs. (i) and (ii) we get,

$$1 \times \text{Eq. (i)} \quad 20x + 50y = 12450$$

$$20x + 20y = 6480$$

$$20x \text{ Eq. (ii)}$$

$$\underline{\hspace{1cm}} \quad \underline{\hspace{1cm}} \quad \underline{\hspace{1cm}}$$

$$30y = 5970$$

$$y = 199$$

$$\therefore x = 125$$

$\therefore$ Number of Rs. 20 notes is 125

$\therefore$ The correct answer is option (c).

58. (a) Given,

$$X = 1! + 2! + 3! + \dots + 100!$$

$$\text{Now, } \frac{X}{24} = \left(\frac{1!+2!+3!+\dots+100!}{24} \right)$$

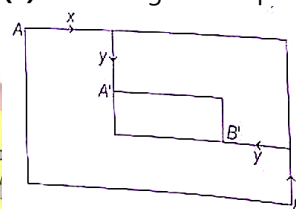
$$\Rightarrow \frac{X}{24} = \left(\frac{1 + 2 \times 1 + 3 \times 2 \times 1 + \dots + 100 \times 99 \times \dots \times 1}{24} \right)$$

$$\Rightarrow \frac{X}{24} = \left(\frac{1 + 2 + 6 + 24 + 120 + \dots + 100 \times 99 \times \dots \times 1}{24} \right)$$

$$\Rightarrow \frac{X}{24} = \frac{9}{24}$$

$\therefore$ Required remainder = 9

59. (a) According to the question,



$$2(A + B) = 100$$

$$\text{And } 2(A' + B') = 40$$

$$A + B = 50$$

$$\text{And } A' + B' = 20$$

$$\text{Now, } A + B = 50$$

$$\Rightarrow (x + y + A' + x + y + B') = 50$$

$$\Rightarrow 2x + 2y + A' + B' = 50$$

$$\Rightarrow 2x + 2y + 20 = 50$$

$$\Rightarrow 2(x + y) = 30$$

$$\text{So } x + y = 15$$

Hence, A walked 15m of distance.

60. (a) According to the venn diagram, Girls who are plays but not a coach is represented by the number common to only girl's and player's circle, which is 16.

61. (c)

Pole length 21 meter

3 meter 2 meter 5 meter 4 meter

1 min 2 min 3 min 4 min

